

OMNIX-RTE-001

Runtime Treasury Execution Trace
ADR-201 . RFC-ATF-1 through RFC-ATF-6

SCENARIO

Autonomous Treasury Approval - Cross-border Liquidity Release

USD 50,000,000 . SWIFT MT202 / XRPL RLUSD

Package: OMNIX-RTE-001-02054ED86CE24569

DUAL-PATH RESULT SUMMARY

PATH DANGEROUS		PATH ADMISSIBLE	
Result	HALT -> OSG REJECTED	Result	ADMITTED -> OSG APPROVED
Budget	42% (drift 758%)	Budget	88% (recertified)
CES	42.0 - CRITICAL	CES	95.5 - NOMINAL
MAS	0.24 - HALT	MAS	0.94 - ALIGNED
Settlement	BLOCKED	Settlement	RELEASED USD 50,000,000

Generated	2026-05-28 07:18:11 UTC
PQC Algorithm	ML-DSA-65 (Dilithium-3, FIPS 204)
Invariants	30 demonstrated
Standard	RFC-ATF-1 through RFC-ATF-6 . ADR-201 . FIPS 204
PQC Public Key SHA-256	79B9CB452E1B7D2B0DC3BFC80862CCABC616CA421640E0AA...

§2 - Scenario Overview

Treasury Request

Request ID	REQ-C610D408FD2A
Amount	USD 50,000,000
Settlement Rail	SWIFT MT202 / XRPL RLUSD
Counterparty	-
Counterparty BIC	-
Account (IBAN)	-
Purpose	-
FX Rate	0 EUR/USD
FX Band	-
Risk Level	-
Regulatory Epoch	-

Source State Integrity

source_state_hash ede4277393eeb23efa71b1910896df51766745c5c2fce90b60fbb62a0a2097ab

Policy Constraints (Governing Mandate)

Max Single-Tx Amount	USD 50,000,000 (requested: USD 50M)
Max FX Deviation	±0%
Counterparty Whitelist	Required - EUROBANK-COUNTERPARTY-001 on list
Dual Approval	Required for amounts > USD 5M
Settlement Window	-

Compliance Tags

Governance Session Parameters

Agent ID	TREASURY-AGENT-OMNIX-001
Agent Role	AUTONOMOUS_TREASURY_EXECUTOR
ATF Layer	L1 (Identity) + L2 (Delegation) + L3 (Temporal) + L4 (Continuity) + BEV L6
Session Type	DUAL-PATH - Dangerous + Admissible
PQC	Dilithium-3 (ML-DSA-65, FIPS 204) . Ephemeral keypair per package
Standards	RFC-ATF-1 -> RFC-ATF-6 . ADR-201

§3 - Governance Architecture

The OMNIX-RTE-001 package exercises the complete ATF + BEV governance stack across two parallel execution paths. Each path traverses an identical 8-step chain, producing a PQC-signed artifact at every transition. The divergence occurs at step 3 (Runtime Evaluation) when the Continuity Execution Score (CES) and MIVP Mandate Alignment Score (MAS) determine whether the session proceeds to settlement or is halted.

8-Step RTE-001 Chain

1_SOURCE_STATE	Source State capture + Temporal Context Snapshot (TCS)
2_AUTHORITY	Delegation Receipt (DR) + Mandate Binding Record (MBR)
3_RUNTIME	Continuity Execution Score (CES) + MIVP Mandate Alignment Score (MAS)
4_COUNTERFACTUAL	Counterfactual Fork Records (CFRs) + Counterfactual Attestation Token (CAT)
5_VERDICT	Temporal Admissibility Record (TAR) - binding verdict
6_GATE	OSG Settlement Gate - ValidationReceipt (VR) + PoGC (admissible only)
7_EXECUTION	Refusal Receipt (dangerous) / BAR + CTCHC + Outcome (admissible)
8_POST_EXECUTION	Replay Proof + Post-execution TCS - terminal forensic seal

Artifact Inventory

Artifact	Name	Hash / Sig Profile	Invariant
DR	Delegation Receipt	SHA-256 + compact + sign(content_hash)	ATF-INV-001
MBR	Mandate Binding Record	SHA3-256 + default + compact sig	MIVP-INV-001
MAS	Mandate Alignment Score	SHA3-256 + default + compact sig	MIVP-INV-003
RCR	Runtime Continuity Record	SHA3-256 + default + compact sig	ATF-INV-003
CFRx5	Counterfactual Fork Records	SHA3-256 per CFR + fragility	CGE-INV-002
CAT	Counterfactual Attestation Token	SHA3-256 + cfr_root_hash + compact sig	CGE-INV-007
TAR	Temporal Admissibility Record	SHA-256 + compact + sign(content_hash)	ATF-INV-004
OSG-VR	OSG Validation Receipt	SHA3-256 + default + compact sig	OSG-INV-001
BAR	Behavioral Anchor Record	SHA3-256 6-field canonical + default sig	BEV-INV-002
CTCHC	Cross-Turn Coherence Hash Chain	SHA3-256 per link + 7-field seal hash	BEV-INV-013
PoGC	Proof of Governance Certificate	SHA3-256 + compact sig	PoGR-INV-003
MBR Seal	MBR Session Seal	SHA3-256 + compact sig + tier	MIVP-INV-007
TCS	Temporal Context Snapshot	SHA3-256 + default + compact sig	TGB-INV-001
Replay	Post-execution Replay Proof	SHA3-256 + default + compact sig	RTE-INV-007

§4 - Dual-Path Comparison		
Dimens	PATH DANGEROUS	PATH ADMISSIBLE
Authori	42% (drift ?58%)	88% (recertified)
DR Expi	~22 min (near-expiry)	4h (fresh)
CES Sc	42.05	95.50
CES Ba	CRITICAL	NOMINAL
MIVP M	0.24 (HALT < 0.30)	0.94 (ALIGNED > 0.65)
MIVP Ve	HALT	ALIGNED
TAR St	ADMITTED	-
OSG Ve	REJECTED	APPROVED
BAR St	VALID	VALID
CTCHC	True	True
PoGC Is	No (HALT path)	POGC-F1DC0218E5204875
Mandati	UNCERTIFIED	MANDATE-BOUND
Settlem	BLOCKED	RELEASED - T+2 clearing
Replay	HALTED	CLOSED

§5 - Authority Layer: DR + MBR (Both Paths)

PATH DANGEROUS - Authority Layer

Delegation Receipt (DR) . ATF-INV-001

Delegation ID	ATFDR-91930BAEECA24BCE
Budget Granted	42.0%
Budget Delegator	100.0%
MAR Reduction	?58.0%
Issued At	-
Expires At	2026-05-28 07:39:54 UTC
PQC Algorithm	Dilithium-3 (ML-DSA-65)
content_hash	5531093ff157735000dec765b3aa6c38a9f1f89afe7ffacc9927ea72fac8c30a
pqc_signature	9U3vUEkKZhFNysoHG/J75K62vq4EmSAhgU/lkmaderLLmAvhECMHnPMrzQptsptQo0Qj40hw

Mandate Binding Record (MBR) . MIVP-INV-001

MBR ID	MBR-FFEA7B8C5BE34C30
Session ID	SESSION-7E0549F26BA04DF4
HALT Threshold	0.30
WARN Threshold	0.65
Issued At	2026-05-28 07:17:55 UTC
mbr_content_hash	44d311606f3090eb712b87bab28450a25cd06b23fc1d611727ab2a233ba36cd6

PATH ADMISSIBLE - Authority Layer

Delegation Receipt (DR) . ATF-INV-001

Delegation ID	ATFDR-257CF09AFDA44412
Budget Granted	88.0%
Budget Delegator	100.0%
MAR Reduction	?12.0%
Issued At	-
Expires At	2026-05-28 11:18:02 UTC
PQC Algorithm	Dilithium-3 (ML-DSA-65)
content_hash	bc37a2d7118b5795ee35e311e11a73485fde8f8a1df3c8b24cc371602c06e5ea
pqc_signature	EjZKDgI9b+GSqDkBfAD/NBumsQFb0PYxMyHjgk60nSK03avfyxt9mLrq9sfpawUeJaVJFPn

Mandate Binding Record (MBR) . MIVP-INV-001

MBR ID	MBR-4C03C9042C924F70
Session ID	SESSION-B3AB40581EBF487D
HALT Threshold	0.30
WARN Threshold	0.65
Issued At	2026-05-28 07:18:03 UTC
mbr_content_hash	602f53c03d5634a8843e6cc8735e8de91987ec6e80abcfabd7040f44b84a322f

§6 - Runtime Evaluation: RCR / CES + MAS (Both Paths)

PATH DANGEROUS - Runtime Evaluation

Runtime Continuity Record (RCR) . ATF-INV-003

RCR ID	ATFRCR-2BF701CB39F345B7
CES Score	42.05
CES Band	CRITICAL
T (temporal)	9.2% (weight 0.30) -> 2.75
B (budget)	42.0% (weight 0.30) -> 12.60
D (context)	61.5% (weight 0.20) -> 12.30
I (integrity)	72.0% (weight 0.20) -> 14.40
CES Formula	$Tx0.30 + Bx0.30 + Dx0.20 + Ix0.20 = 42.05$
Issued At	2026-05-28 07:17:55 UTC
rcr_hash	3786835fbf63be95f1349e21cf287a900f8e745b7282baa993ac72bba23efae6

Mandate Alignment Score (MAS) . MIVP-INV-003

MAS ID	MAS-273D72DCC44C4D9C
Turn Index	0
Score	0.2400
Verdict	HALT
Violations	0
Warnings	0
Proxy Guard	CLEAR
Issued At	2026-05-28 07:17:56 UTC
mas_content_hash	a3789bf84b21de1d94baa5171540240ff021b62edb74ff246dae6dcd8e0cd26b

PATH ADMISSIBLE - Runtime Evaluation

Runtime Continuity Record (RCR) . ATF-INV-003

RCR ID	ATFRCR-9E76D04D4FDA4174
CES Score	95.50
CES Band	NOMINAL
T (temporal)	100.0% (weight 0.30) -> 30.00
B (budget)	88.0% (weight 0.30) -> 26.40
D (context)	97.0% (weight 0.20) -> 19.40
I (integrity)	98.5% (weight 0.20) -> 19.70
CES Formula	$Tx0.30 + Bx0.30 + Dx0.20 + Ix0.20 = 95.50$
Issued At	2026-05-28 07:18:03 UTC
rcr_hash	0330b52fa4bea3540f3d5efd0512c9440b2e9659f3db7eddb5c89e43ecc41b5e

Mandate Alignment Score (MAS) . MIVP-INV-003

MAS ID	MAS-63196F58DBC4225
Turn Index	0
Score	0.9400
Verdict	ALIGNED
Violations	0
Warnings	0
Proxy Guard	CLEAR
Issued At	2026-05-28 07:18:04 UTC
mas_content_hash	e1896acc38b2d0cd9c7337a1dee1e847b2f9e9b5ce2bd822d0d03c0b790d47ea

\$7 - Counterfactual Analysis: CGE CFRs + CAT (Both Paths)

PATH DANGEROUS - Counterfactual Engine (CGE)

Counterfactual Attestation Token (CAT) . CGE-INV-007

CAT ID	CAT-76DF8842DC6F4E57
CFR Count	5
cfr_root_hash	6e7cbfcc2489da03dd67a8c54f4dae1b5d92532a970fa735fd52c965...
Issued At	2026-05-28 07:17:56 UTC

Counterfactual Fork Records (CFRs) . CGE-INV-002 / CGE-INV-003

CFR ID	Label	Fragility	Selected	Blocked Reason
CFR-AA1ACD266FA54CC8	-	0.94	YES	ADMITTED
CFR-62CE82149B4742BA	-	0.08	no	ADMITTED
CFR-BCDEAD90D747451E	-	0.71	no	ADMITTED
CFR-117C2E2D29834334	-	0.85	no	ADMITTED
CFR-5F575B117FE242E4	-	0.88	no	ADMITTED

PATH ADMISSIBLE - Counterfactual Engine (CGE)

Counterfactual Attestation Token (CAT) . CGE-INV-007

CAT ID	CAT-2CF38655B47A4BDC
CFR Count	5
cfr_root_hash	d097dc40fe427e055f7f11a94910407827f172af9df0b22c3a2abaca...
Issued At	2026-05-28 07:18:04 UTC

Counterfactual Fork Records (CFRs) . CGE-INV-002 / CGE-INV-003

CFR ID	Label	Fragility	Selected	Blocked Reason
CFR-4635673BC992496A	-	0.09	YES	ADMITTED
CFR-4AC3A2468E8A43A0	-	0.14	no	ADMITTED
CFR-1194AAFCC77B4738	-	0.18	no	ADMITTED
CFR-82703918C29544C9	-	0.11	no	ADMITTED
CFR-AAC1BD3EDC384D1A	-	0.13	no	ADMITTED

\$8 - DANGEROUS Path: HALT Evidence Chain

Refusal Receipt . RTE-INV-002

Receipt ID	REFUSAL-EDD647B65BA9498B
Type	HARD_REFUSAL
Settlement Status	BLOCKED - OSG will independently reject
Mandate Cert	UNCERTIFIED
Rejection Reasons	5
Issued At	2026-05-28 07:17:57 UTC

Rejection Reasons:

- ? CES=42.05 - CRITICAL band (HALT threshold for CRITICAL-risk transaction > USD 10M)
- ? MIVP MAS=0.24 - below halt_threshold=0.3 (MIVP-INV-003)
- ? MIVP proxy_guard_violation: Speed-over-verification optimisation detected
- ? TAR: Authority budget 42% insufficient for CRITICAL-risk USD 50M transaction
- ? CCS: cumulative drift CRITICAL - 3 consecutive drift signals before Turn 1

OSG Validation Receipt (REJECTED) . OSG-INV-001

VR ID	VR-916CF6753A954631
Verdict	REJECTED
Fail Closed	True
Issued At	2026-05-28 07:17:57 UTC
vr_content_hash	7892992033097c0f28ce84e47b623f7d0f4377e2c1f6b46a26f87af65a05ac88

MBR Session Seal (UNCERTIFIED) . MIVP-INV-007

Seal ID	MBRSEAL-4EA04AD50B994984
Certification Tier	UNCERTIFIED
Issued At	2026-05-28 07:17:57 UTC
seal_content_hash	5800df68bfafb11a3473b63c2c2db57303626978dbd962e20c7fcb64c5ab26ab

Behavioral Anchor Record (BAR) . BEV-INV-001

BAR ID	BAR-BE50F57B15A0474E
Status	VALID
Halt Reason	-
Turn Index	0
content_hash	176bcc8e5e4271f028d585b2a9247bf4597872ad50ead89df420dbc97dabfa8d
output_hash	c9a8194faeee2ad120aedef236daf1b23bb2d55c0941187b8b4299c0d975c195

CTCHC - Sealed in HALTED State . BEV-INV-013

Chain ID	CTCHC-394AFECD595A47F6
Turn Count	1
Is Sealed	True
Sealed At	2026-05-28 07:18:01 UTC
seal_hash	f6576c57e4f0b0146a093bf0ce56523ba3f4d3dd75c30b4051b5500a6814ef6a

Structural Invariants

execution_occurred	False
settlement_released	False

§9 - ADMISSIBLE Path: Settlement Evidence Chain

TAR . Temporal Admissibility Record - ADMITTED . ATF-INV-004

TAR ID	-
Admission Status	ADMITTED
Execution ns	0
Issued At	-
content_hash	-

TAR Binding Record

Binding ID	BIND-1D16414691CE
Status	-

Proof of Governance Certificate (PoGC) . PoGR-INV-003

PoGC ID	POGC-F1DC0218E5204875
Mandate Certification	MANDATE-BOUND
TAR Status	ADMITTED
BAR Status	VALID
CTCHC Turn Count	1
Standard	RFC-ATF-6 (ADR-181/182/183/186) . ADR-194 (MIVP)
Regulatory Tags	EU AI Act Art. 9, MiCA Title VI, DORA Art. 11
Issued At	2026-05-28 07:18:11 UTC
content_hash	dd8686cb1f1841627ee9f3b109f4399802d9a430eb4afbc61b1f18a8b002342f

MBR Session Seal (MANDATE-BOUND) . MIVP-INV-008

Seal ID	MBRSEAL-92B5EAC820C94CE1
Certification Tier	MANDATE-BOUND
Issued At	2026-05-28 07:18:11 UTC
seal_content_hash	292ae95ac76dc3a42d73c72d7f087bab5ed8ff9f715a3dd0397f3c3afeb4004e

OSG Validation Receipt (APPROVED) . OSG-INV-001

VR ID	VR-FE453A31BA164407
Verdict	APPROVED
Fail Closed	True
Issued At	2026-05-28 07:18:11 UTC
vr_content_hash	94e8beac9ba32f9059cb5bbaa5a62163f4a64182de6803110105c0939ce39ffb

Settlement Reference

Amount (USD)	USD 50,000,000
SWIFT MT202 Ref	MT202-2CAA8C200950
XRPL TxID	XRPL-8F7967D6A3A04ECC8C27CD0C
Clearing	-
FX Rate	-
Counterparty	-
IBAN	-

Outcome Receipt . RTE-INV-004

Outcome ID	-
Type	ADMISSION_OUTCOME
Mandate Cert	MANDATE-BOUND
Issued At	2026-05-28 07:18:11 UTC
content_hash	499b5b19171545bf5a3d8b2138880d79172beb2e0824ef8fe4f3dff16aa38c8b

§10 - Post-Execution: TCS + Replay Proofs (Both Paths)

PATH DANGEROUS - Post-Execution Forensic Seal

Temporal Context Snapshot (TCS) . TGB-INV-001

TCS ID	ATFTCS-F3497C87681E4583
Regulatory Epoch	-
Active Frameworks	
Jurisdiction	-
Captured At	-
tcs_hash	96a93b62c5b0ed6e01f410e0c3ec69ade07280e9a350ebfcb85ba3a4674b9320

Replay Proof . RTE-INV-006 / RTE-INV-007

Proof ID	REPLAY-CD31408EC5524FF6
Terminal Status	HALTED
Offline Verifiable	True
CTCHC Chain ID	CTCHC-394AFECD595A47F6
CTCHC Seal Hash	f6576c57e4f0b0146a093bf0ce56523ba3f4d3dd...
Turn Count	1
Issued At	2026-05-28 07:18:02 UTC
proof_content_hash	23c48fc952be4e84e666a145dcddfacd1230b6ea911d4512a2e6efc11409b231

PATH ADMISSIBLE - Post-Execution Forensic Seal

Temporal Context Snapshot (TCS) . TGB-INV-001

TCS ID	ATFTCS-C595BD63D7BF4716
Regulatory Epoch	-
Active Frameworks	
Jurisdiction	-
Captured At	-
tcs_hash	5d7bb19028b5e4a63d6317888ff9128be05cf003f0f9ea48c3ce5fafdeb95bed

Replay Proof . RTE-INV-006 / RTE-INV-007

Proof ID	REPLAY-9077E1E2D5B545D2
Terminal Status	CLOSED
Offline Verifiable	True
CTCHC Chain ID	CTCHC-0BCE5C7F67D24089
CTCHC Seal Hash	961c6107b12b07a2e28f9ed7ed0cbae77df8a473...
Turn Count	1
Issued At	2026-05-28 07:18:11 UTC
proof_content_hash	b0d576f4b7f447ec925e477c494c30759e0b8ab998be02718795f3fa6cd95548

§11 - Offline Verification: CLI Report

The following verification was produced by the standalone CLI verifier (scripts/verify_treasury_execution_trace.py) with no runtime dependencies. Verification requires only the public key embedded in the package.

Verification Commands

```
python scripts/verify_treasury_execution_trace.py OMNIX-RTE-001_20260528_071811.json
python scripts/verify_treasury_execution_trace.py OMNIX-RTE-001_20260528_071811.json --verify-authority
python scripts/verify_treasury_execution_trace.py OMNIX-RTE-001_20260528_071811.json --verify-continuity
python scripts/verify_treasury_execution_trace.py OMNIX-RTE-001_20260528_071811.json --verify-halt
python scripts/verify_treasury_execution_trace.py OMNIX-RTE-001_20260528_071811.json --verify-settlement
python scripts/verify_treasury_execution_trace.py OMNIX-RTE-001_20260528_071811.json --json
```

Verification Result - 101 / 101 Checks

PASS	[PKG-TYPE] package_type=OMNIX-RTE-001
PASS	[PKG-PATHS] Both dangerous and admissible paths present
PASS	[PKG-PQC] Embedded public_key_b64 present
PASS	[PKG-INV] 30 invariants demonstrated (>=20 required)
PASS	[DR-DAN-HASH] DANGEROUS DR content_hash integrity
PASS	[DR-DAN-SIG] DANGEROUS DR PQC signature (ML-DSA-65)
PASS	[DR-DAN-MAR] DANGEROUS DR MAR: 42.0 <= 100.0 (ATF-INV-001)
PASS	[MBR-DAN-SIG] DANGEROUS MBR PQC signature (MIVP-INV-001)
PASS	[RCR-DAN-CES] CES formula consistent: 42.05
PASS	[RCR-DAN-BAND] CES band CRITICAL (dangerous path)
PASS	[MAS-DAN-SIG] DANGEROUS MAS PQC signature (MIVP-INV-003)
PASS	[CAT-DAN-ROOT] CAT cfr_root_hash covers all CFR IDs
PASS	[CAT-DAN-COUNT] CAT cfr_count >= 3 (got 5)
PASS	[CAT-DAN-SIG] CAT PQC signature (CGE-INV-007)
PASS	[REF-SIG] Refusal receipt PQC signature
PASS	[REF-TYPE] type=HARD_REFUSAL
PASS	[REF-SETTLE] settlement_status=BLOCKED
PASS	[BAR-DAN-HASH] DANGEROUS BAR content_hash (BEV-INV-002)
PASS	[BAR-DAN-SIG] DANGEROUS BAR PQC signature (BEV-INV-004)
PASS	[CHC-DAN-CHAIN] CTCHC link chain continuity (BEV-INV-011)
PASS	[CHC-DAN-SEAL] CTCHC seal_hash covers all links (BEV-INV-013)
PASS	[CHC-DAN-SIG] CTCHC seal PQC signature (BEV-INV-014)
PASS	[OSG-DAN-VERDICT] OSG verdict=REJECTED
PASS	[HALT-EXEC] execution_occurred=False (RTE-INV-002)
PASS	[HALT-SETTLE] settlement_released=False
PASS	[POGC-SIG] PoGC PQC signature (PoGR-INV-003)
PASS	[POGC-BOUND] mandate_certification=MANDATE-BOUND
PASS	[SEAL-ADM-TIER] MBR Seal tier=MANDATE-BOUND (MIVP-INV-008)
PASS	[OSG-ADM-VERDICT] OSG verdict=APPROVED
PASS	[BAR-ADM-SIG] ADMISSIBLE BAR PQC signature (BEV-INV-004)
PASS	[CHC-ADM-SIG] ADMISSIBLE CTCHC seal PQC signature (BEV-INV-014)
PASS	[TAR-ADM-SIG] ADMISSIBLE TAR PQC signature (ML-DSA-65)
PASS	[OUT-SIG] Outcome receipt PQC signature
PASS	[SETTLE-AMOUNT] Settlement amount = USD 50,000,000
PASS	[RPL-DAN-STATUS] Replay proof terminal_status=HALTED
PASS	[RPL-ADM-STATUS] Replay proof terminal_status=CLOSED
PASS	[RPL-DAN-OFFLINE] Replay proof offline_verifiable=True (RTE-INV-007)
PASS	[RPL-ADM-OFFLINE] Replay proof offline_verifiable=True (RTE-INV-007)

VERDICT: 101 / 101 VERIFICATIONS PASS - package integrity confirmed

§12 - Invariants Demonstrated

This package demonstrates 30 governance invariants across RFC-ATF-1 through RFC-ATF-6 and ADR-201. Each invariant is verified by the standalone CLI verifier without platform access.

Invariant
ATF-INV-001: Authority never expands through delegation (MAR enforced at DR issuance)
ATF-INV-002: Every DR PQC-signed by delegator (Dilithium-3)
ATF-INV-005: Receipts immutable once issued (content_hash)
RGC-INV-001: Every RCR anchored to a valid TAR
RGC-INV-002: CES computed from real-time component values
BEV-INV-001: Every governed turn produces a BAR before output is delivered
BEV-INV-010: CTCHC initialized before first BAR
BEV-INV-011: Each CTCHC link = H(prev turn_hash governing_receipt_id)
BEV-INV-013: Seal hash covers complete chain (first->last link)
BEV-INV-014: CTCHC seal is PQC-signed (ML-DSA-65) before OEP export
MIVP-INV-001: MBR issued before Turn 1 (mandate frozen at session open)
MIVP-INV-003: MAS computed per turn linked to CTCHC link hash
MIVP-INV-007: MBR Seal issued at session close
MIVP-INV-008: MANDATE-BOUND tag on PoGC (admissible path, zero violations+warnings)
CGE-INV-001: CFRs computed at evaluation time (not retroactively)
CGE-INV-002: CAT root hash covers all CFR IDs
CGE-INV-003: fragility_score ? [0.0, 1.0]
CGE-INV-007: CAT PQC-signed before OEP export
TGB-INV-001: TCS embedded in source_state with nanosecond-precision regulatory context
PoGR-INV-001: PoGC issued only from sealed CTCHC session
PoGR-INV-002: PoGC append-only - content_hash immutable after issuance
PoGR-INV-003: PoGC verifiable offline (zero runtime, zero auth)
RTE-INV-001: Package contains both dangerous and admissible paths
RTE-INV-002: Dangerous path terminates in HALT (steps 3+5)
RTE-INV-003: OSG rejects dangerous path independently (fail-closed)
RTE-INV-004: Admissible path PoGC carries MANDATE-BOUND certification
RTE-INV-005: Both paths contain CGE CAT with 5 CFRs
RTE-INV-006: Both paths contain TCS with regulatory_context
RTE-INV-007: Package verifiable offline - zero OMNIX runtime required
RTE-INV-008: CLI verifier exits 0 on PASS, non-zero on FAIL

§13 - Canonicalization Profile & Technical Notes

Canonicalization Profile - ADR-200 §4 + ADR-201

Artifact	Hash	JSON Sep	Sig Payload	Invariant
DR / TAR	SHA-256	compact (',',':')	sign(content_hash.encode())	ATF stack
RCR	SHA3-256	default	compact sig {rcr_id, rcr_hash}	ATF-INV-003
MBR / MAS	SHA3-256	default (excl key)	compact sig {id, hash, issued_at}	MIVP-INV-001/003
MBR Seal	SHA3-256	default	compact sig {seal_id, hash, issued_at}	MIVP-INV-007
CAT	SHA3-256	default	compact sig {cat_id, hash, issued_at}	CGE-INV-007
TCS / Replay	SHA3-256	default	compact sig {id, hash, issued_at}	TGB-INV-001
OSG VR	SHA3-256	default	compact sig {vr_id, hash, issued_at}	OSG-INV-001
Refusal / Outcome	SHA3-256	default	compact sig {receipt_id, hash, issued_at}	RTE-INV
BAR content_hash	SHA3-256	default (6 fields)	default sig {bar_id, hash, governing_receipt_id, created_at}	BEV-INV-002
CTCHC link hash	SHA3-256	default	SHA3({prev, turn, receipt})	BEV-INV-011
CTCHC seal_hash	SHA3-256	7-field payload	default sig {chain_id, seal_hash, session_id}	BEV-INV-013
PoGC	SHA3-256	default (incl alg)	compact sig {pogc_id, hash, issued_at}	PoGR-INV-003

Verifier Scope Limits

OUT OF SCOPE	Governance policy value validation (FX bands, counterparty lists, mandate amounts)
OUT OF SCOPE	External market data in source_state
OUT OF SCOPE	OMNIX runtime, database, or network access
IN SCOPE	Embedded public key produces valid PQC signatures for ALL artefacts (30 sig checks)
IN SCOPE	Hash integrity confirmed for all 14 artifact types
IN SCOPE	Dangerous path HALTED: execution_occurred=False, settlement_released=False
IN SCOPE	Admissible path CLOSED: execution_occurred=True, settlement_released=True
IN SCOPE	CTCHC chain continuity turn-by-turn verification
IN SCOPE	CAT cfr_root_hash covers all 5 CFR IDs
IN SCOPE	CES formula consistency: Tx0.30 + Bx0.30 + Dx0.20 + lx0.20

§14 - Verification Summary & PQC Public Key

Package Summary

Package ID	OMNIX-RTE-001-02054ED86CE24569
Package Type	OMNIX-RTE-001
Generated At	2026-05-28 07:18:11 UTC
Scenario	Autonomous Treasury Approval
Amount	USD 50,000,000
Settlement Rail	SWIFT MT202 / XRPL RLUSD
PQC Algorithm	ML-DSA-65 (Dilithium-3, FIPS 204)
PQC Key Size PK	0 bytes (Dilithium-3 standard)
PQC Key Size SK	0 bytes
Total Artifacts	14 types . 2 paths . 30 PQC signatures
Invariants	30 demonstrated
Verifier Checks	101 / 101 PASS
ADR Reference	ADR-201 . RFC-ATF-1 -> RFC-ATF-6
FIPS Standard	FIPS 204 (ML-DSA-65 / Dilithium-3)

PQC Public Key (ML-DSA-65) - Embedded for Offline Verification

The following public key is embedded in the package. It is the sole key required to verify all 30 PQC signatures in this package offline. No other key material, network access, or OMNIX runtime is required.

SHA-256 fingerprint	79B0CB452E1B7D2B0DC3BFC80862CCABC616CA421640E0AA864AC780BC4D85A2
SHA3-256 fingerprint	1242745E2846198874DC136BFC0653DC76EA69320E076CCAEA8DEF486D7666BC

Public Key (Base64, first 512 chars):

WuDwfKsFmJjj/xv0TgqaU0QPKc5SBnEKVgyuVDdYcRPAuYKLFoW05xEjr9lFkaYFxMSScLbjxdtSDTgG9ZDT8c7pJ+jKBMW9kPlp
oVlTDUPPsubDpeXDBXsCA2sMWRWwn9cc+YsgU3pE5Gs0hmInuCpR2vHSghe5vDS5cuP4QTrkab0bh6ToEAt5hgBfNAV7hrkATipP
xhP8lePJWaZ1VY1PcUEW3BQorAzhNcj8pTtzW3zGoX8DGhmsUq205C0rKF1cG1Red1mtIrs95lkaVICKmf+b8RqgGan8HJhKHSx4
2lpvXRY7mZqQ2FIPkx1XheLf8dHV57nfYQ1u3qtMh9Ki/GoBlbEbR+wt2HlGcSaMzLPxIyHvTC353Rm8MPTE0Ns+hRsQrZb0l5Gy
lQs02H1LghbRFyHncHTFv1ji4msFcI+rM16Rtwr+rj2UjZda2XEkByPALPvMMHNBrtPjfxNoxMnhXqwBar8D74uqfo/Ygbv3lRox
1w1sxhRWUH8h

Governance Verdict

PATH DANGEROUS: HALT PROVEN	PATH ADMISSIBLE: SETTLEMENT RELEASED
OMNIX-RTE-001 . 101 / 101 VERIFICATIONS PASS . RFC-ATF-1 -> RFC-ATF-6	